PHY201 Practice Exam on chapter(s): 9, 10

Name:____

These questions assess your understanding of the material from Chapters 9, 10 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. Julia, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 4.0 cm from the reel's axis of rotation. The reel is given an angular acceleration of 127 rad/s^2 for 2.4 s. At what speed is the fishing line leaving the reel after 2.4 s elapses?

ANSWER: __________________________

2. Ryan is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. He exerts a force of 179 N through a rotation of 1.37 rad (78°). He exerts the force perpendicular to the grindstone's 25-cm radius and the effects of friction are negligible. What is the grindstone's final angular velocity if its mass is 132 kg? Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: __________________________

3. Nathan begins spinning a large grindstone from rest by placing his hand on its edge and exerting a force through part of a revolution. He exerts a force of 353 N through a rotation of 0.21 rad (12°). He exerts the force perpendicular to the grindstone's 32-cm radius and the effects of friction are negligible. If the grindstone's mass is 39 kg, what is its final rotational kinetic energy? Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: __________________________

4. Two children, Allison and Michael, and a toy hippopotamus are balancing on a long seesaw of negligible mass. Allison has a mass of 22 kg and sits 2.5 m away from the fulcrum. Michael, who has a mass of 15 kg, sits on the other side of the fulcrum, opposite to Allison. The toy hippopotamus has a mass of 3.6 kg and is on Michael's side of the seesaw, a distance of 0.8 m away from the fulcrum. What is Michael's distance from the fulcrum?

ANSWER: __________________________

5. Rachel, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 4.4 cm from the reel's axis of rotation. The reel is given an angular acceleration of 117 rad/s^2 for 1.7 s. How many revolutions does the reel make?

ANSWER: __________________________

6. Grace places a ladder of mass 13 kg against her house. The feet of the ladder are resting on her concrete patio at a distance of 5.3 m from the house. The top of the ladder rests against a plastic rain gutter. The friction between the ladder and gutter is negligible. The top of the ladder is at a height of 5.7 m above the concrete patio. The ladder's center of gravity is at its midpoint. Grace is standing on a rung at a horizontal distance of 3.6 m from the house. If Grace's mass is 67 kg, what is the minimum static-friction coefficient that would prevent the feet of the ladder from slipping on the concrete?

ANSWER: __________________________

7. Emma places an outdoor sandwich-board advertising sign on the ground. The sign has a taut chain 0.37 m vertically below its hinge. The entire sign's mass is 18 kg. The sign has a vertical height of 0.92 m. The hinge is opened so that the feet of the sign measure 1.4 m apart. Calculate the magnitude of the tension force that the chain exerts on each of the sign's sides, assuming there is no friction between the feet and the sidewalk. (Drawing tip: the sign resembles a large A. The chain resembles the horizontal line in the A and the hinge is at the top of the A.)

ANSWER: __________________________

8. David is designing a set to be shown in a play. A 76-kg actor stands on a 95-kg platform that is suspended by two ropes, one on its left side, and another on its right side. The left rope is attached to a weight of mass m_L through a frictionless pulley; the right rope is attached to a weight of mass m_R through another frictionless pulley. When the weights are released, the platform must descend with an acceleration of 0.5 m/s^2 without rotating. David needs to make the platform 2.4-m long and the actor must stand 0.2-m from the left side. What must m_L be?

ANSWER: ______________________

9. Mary flips her bicycle upside down and moves the pedals so that the rear wheel begins spinning. If the wheel starts from rest and reaches a final angular velocity of 309 rpm in 4.1 s, calculate the angular acceleration in rad/s^2.

ANSWER: ______________________

10. Anna is designing a new 3.1-m-long, 20-kg diving board for DiveMaster, Inc. She bolted a prototype into the ground 2.8 m away from the pool-side end of the diving board with its spring being 1.8 m away from the pool-side end. Anna notes that the spring compresses by 9 cm at equilibrium under the weight of the diving board alone. When Anna, who has a mass of 75 kg, gets on the prototype diving board at a distance of 1.1 m from the pool-side end, by how much does the spring compress at equilibrium? Report your answer in centimeters.

ANSWER: ______________________

11. Stephen places an outdoor sandwich-board advertising sign on the ground. The sign used to have a chain 0.61 m vertically below its hinge, but the chain is broken. The sign's mass is 16 kg. The sign has a vertical height of 1.48 m. The hinge is opened so that the feet of the sign measure 2.7 m apart. What is the minimum coefficient of static friction between the feet of the sign and the sidewalk? To draw the sign, you may draw a large A. The chain, now broken, would have resembled the horizontal line in the A and the hinge is at the top of the A.

ANSWER: ______________________

12. A powerful motorcycle can accelerate from rest to 31.8 m/s in 3.2 s. What is the angular acceleration of its 0.35-m-radius wheels?

ANSWER: ______________________

13. A typical small rescue helicopter has four blades, each of which is 4.0-m long and has a mass of 45 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 1050 kg. If the helicopter's blades rotate at 299 rpm on the ground, to what height could the helicopter rise if all of its blades' rotational kinetic energy is used to lift it?

ANSWER: ______________________

14. Leah, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 5.1 cm from the reel's axis of rotation. The reel is given an angular acceleration of 86 rad/s^2 for 3.0 s. How many meters of fishing line come off the reel in this time?

ANSWER: ______________________

15. Charles is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. How much work does he do if he exerts a force of 330 N through a rotation of 0.32 rad (18°)? He exerts the force perpendicular to the grindstone's 49-cm radius and the effects of friction are negligible. Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: ______________________

16. Victoria places a ladder of mass 21 kg against her house. The feet of the ladder are resting on her concrete patio at a distance of 0.9 m from the house. Between the ladder and the concrete, $\mu_s = 0.47$. The top of the ladder rests against a plastic rain gutter. The friction between the ladder and gutter is negligible. The top of the ladder is at a height of 1.31 m above the concrete patio. The ladder's center of gravity is at its midpoint. Victoria is standing on a rung at a horizontal distance of 0.25 m from the house. What is the maximum mass that Victoria could have without causing the ladder to slip on the concrete?

ANSWER: ______________________

17. Chloe places a ladder of mass 22 kg against her house. The feet of the ladder are resting on her concrete patio at a distance of 7.1 m from the house. The top of the ladder rests against a plastic rain gutter. The friction between the ladder and gutter is negligible. The top of the ladder is at a height of 5.5 m above the concrete patio. The ladder's center of gravity is at its midpoint. Chloe is standing on a rung at a horizontal distance of 1.8 m from the house. If Chloe's mass is 88 kg, what is the magnitude of the friction force between the feet of the ladder and the concrete?

ANSWER: ____________________________

18. Paul is building a seesaw for his daughter, Katherine, and plans on making it 16 ft long. He plans to sit on one side, 7.8 ft away from the center of the seesaw, and his daughter will sit on the other side, 7.8 ft away from the center of the seesaw. Paul has weighed himself, his daughter, and the seesaw and learned that his weight is 222 lbs, his daughter's weight is 29 lbs, and the weight of the seesaw is 19.2 lbs. He wants to place the fulcrum so that the seesaw is perfectly balanced, so that Katherine can cause a rotation in the seesaw just as easily as he can. Where must the fulcrum be placed?

ANSWER: ____________________________

19. Lauren is trying to determine the maximum mass that she can safely hold. Her elbow joint is 28 mm from her bicep, 16 cm from the center of mass of her forearm, and 38 cm from the object. What is the maximum mass that Lauren can hold if, when her bicep is at its limit, her upper-arm bone exerts a force of 3000 N on her elbow joint? Note that the angle between her forearm and her upper arm is 90° and that her forearm has a mass of 2700 g.

ANSWER: ____________________________

20. Mary flips her bicycle upside down and moves the pedals so that the rear wheel begins spinning. If the wheel starts from rest and reaches a final angular velocity of 309 rpm in 4.1 s, calculate the angular acceleration in rad/s^2.

ANSWER: ____________________________

21. Brandon wants to place three blocks on a 1.6-kg, 12-m-long wooden board to make it balance atop a triangular wedge of wood. He places a 18-kg purple block a distance of 1.0 m from the left end of the board, a 1.4-kg yellow block a distance of 9.3 m from the right end of the board, and places the board on the wedge at a distance of 2.0 m from the left end of the board. Where must he place a 2.6-kg orange block, measured from the right end of the board, in order to balance the board?

ANSWER: ____________________________

22. Samantha is using a winch and rope to hold up a 6.8-kg water-filled bucket. The winch is made up of a 40-kg cylindrical log with a radius of 28 cm with a light handle extending 53 cm from the center. The rope is attached to the log, wraps around it, and is tied to the bucket. By turning the handle, Samantha would cause the log to rotate, lifting the bucket. What is the magnitude of the force that Samantha must apply to the stationary handle to keep the bucket from moving? The moment of inertia of the cylindrical log is $(1/2)MR^2$.

ANSWER: ____________________________

23. Efe flips his bicycle upside down and moves the pedals so that the rear wheel begins spinning. The wheel starts from rest and reaches a final angular velocity of 340 rpm in 4.5 s. Efe then slams on the brakes to cause an angular acceleration of -30 rad/s^2. How long does it take the wheel to stop?

ANSWER: ____________________________

24. Adam, a construction worker, holds up a plank of wood 3.2-m long. His left hand pushes an end of the plank down with a force F_L, while his right hand pushes the plank up with a force F_R at a distance of 0.9 m from the left hand. The plank is held in equilibrium. The plank has a mass of 6 kg and its center of gravity is located at the middle of the plank. Calculate the magnitude of force F_L.

ANSWER: ____________________________

25. David, a construction worker, holds up a plank of wood 2.7-m long. His left hand pushes an end of the plank down with a force F_L, while his right hand pushes the plank up with a force F_R at a distance of 0.8 m from the left hand. The plank is held in equilibrium. The plank has a mass of 15 kg and its center of gravity is located at the middle of the plank. Calculate the magnitude of force F_R.

ANSWER: ____________________________

26. A typical small rescue helicopter has four blades, each of which is 4.1-m long and has a mass of 48 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 1250 kg. Calculate the rotational kinetic energy that the helicopter possesses when its blades rotate at 281 rpm.

ANSWER: ______________________________

27. Sam, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 4.6 cm from the reel's axis of rotation. The reel is given an angular acceleration of 71 rad/s^2 for 1.8 s. What is the final angular velocity of the reel?

ANSWER: ______________________________

28. Sophia is with her daughter, Grace, at a park. Grace has climbed onto a merry-go-round. Sophia is continuously pushing on the 48-kg merry-go-round with a force of 104 N at its edge, 1.7 m away from the center. Calculate the angular acceleration that is produced in the merry-go-round while 26-kg Grace is 0.85 m away from its center. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: ______________________________

29. Katherine is with her daughter, Anna, at a park. She is continuously pushing on a 29-kg merry-go-round with a force of 50 N at its edge, 1.2 m away from the center. Anna watches excitedly. Calculate the angular acceleration that is produced in the merry-go-round when no one is on it. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: ______________________________

30. Sam is designing a set to be shown in a play. A 73-kg actor stands on a 87-kg platform that is suspended by two ropes, one on its left side, and another on its right side. The left rope is attached to a weight of mass m_L through a frictionless pulley; the right rope is attached to a weight of mass m_R through another frictionless pulley. When the weights are released, the platform must descend with an acceleration of 0.6 m/s^2 without rotating. Sam needs to make the platform 2.0-m long and the actor must stand 1.1-m from the left side. What must m_R be?

ANSWER: ______________________________